Assignment - 4

Code -

```
#include <iostream>
#include <algorithm>
using namespace std;

int main() {
    int n, capacity;

    cout << "Enter number of items: ";
    cin >> n;

    int profit[n], weight[n];

    cout << "Enter profit and weight of each item:\n";
    for (int i = 0; i < n; i++) {
        cin >> profit[i] >> weight[i];
    }

    cout << "Enter knapsack capacity: ";
    cin >> capacity;

    int dp[n + 1][capacity + 1];

    for (int i = 0; i <= n; i++) {
        for (int w = 0; w <= capacity; w++) {
            if (i == 0 || w == 0)
                dp[i][w] = 0;
            else if (weight[i - 1] <= w)
                dp[i][w] = max(
                    profit[i - 1] + dp[i - 1][w - weight[i - 1]],
                    dp[i - 1][w]
                );
            else
                dp[i][w] = dp[i - 1][w];
```